

Dual Spherical Harmonics for 3D Gaussian Splatting: Novel View Synthesis with Dynamic Lighting

Alp Bora Orgun
CVSSP, University of
Surrey
Guildford, United Kingdom
ao00821@surrey.ac.uk

Marco Volino
CVSSP, University of
Surrey
Guildford, United Kingdom
m.volino@surrey.ac.uk

Adrian Hilton
CVSSP, University of
Surrey
Guildford, United Kingdom
a.hilton@surrey.ac.uk

Jean-Yves Guillemaut
CVSSP, University of
Surrey
Guildford, United Kingdom
j.guillemaut@surrey.ac.uk

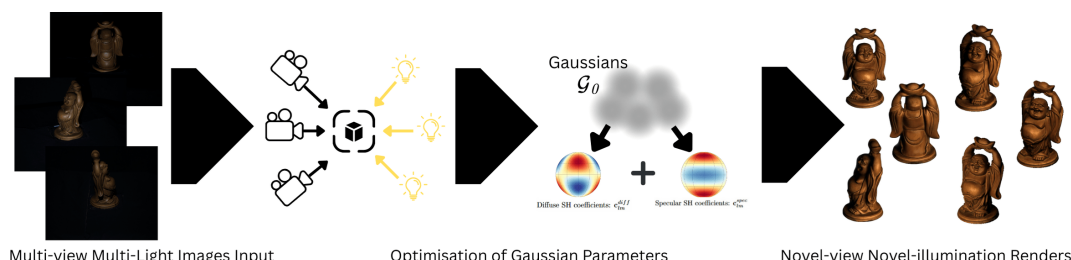


Figure 1: Our approach: from multi-view, multi-light inputs to novel view synthesis under novel illumination.

Abstract

We present a novel framework for scene representation and rendering that combines 3D Gaussian splatting with dual spherical harmonics decomposition to enable high-quality novel view synthesis under arbitrary lighting conditions. Our key innovation lies in separately modeling diffuse and specular reflection components using distinct directional parameterizations: view direction for diffuse components and reflection direction for specular components. This dual spherical harmonic (SH) decomposition, enables realistic rendering of reflective surfaces while maintaining the computational efficiency of Gaussian splatting. To address geometric fidelity challenges, our method incorporates a comprehensive optimization strategy including silhouette-guided geometric constraints for sharp opacity boundaries, HDR saturation loss for handling bright regions in high dynamic range environments, and multi-faceted normal vector optimization with photometric and physics-based supervision. Experimental validation on the DiLiGenT-MV benchmark demonstrates superior performance compared to existing methods, with significant improvements in PSNR, SSIM, and LPIPS metrics while achieving fast optimisation and real-time rendering performance suitable for interactive applications.

CCS Concepts

• **Computing methodologies** → **Machine learning**; **Rasterization**; **Reflectance modeling**; **Rendering**;

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CVMP '25, London, United Kingdom

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-2117-5/25/12

<https://doi.org/10.1145/3756863.3769709>

Keywords

3D Gaussian splatting, novel view synthesis, relighting, spherical harmonics, reflectance modeling, neural rendering, dynamic lighting, silhouette optimization, real-time rendering, computer graphics

ACM Reference Format:

Alp Bora Orgun, Marco Volino, Adrian Hilton, and Jean-Yves Guillemaut. 2025. Dual Spherical Harmonics for 3D Gaussian Splatting: Novel View Synthesis with Dynamic Lighting. In *European Conference on Visual Media Production (CVMP '25)*, December 03–04, 2025, London, United Kingdom. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3756863.3769709>

1 Introduction

The ability to synthesize novel views of a scene under arbitrary lighting conditions is a fundamental challenge in computer vision and computer graphics. This capability is crucial for applications such as virtual reality, augmented reality, and photorealistic rendering. At the heart of this problem lies the need for performant yet expressive scene representations that can accurately capture both the geometry and complex appearance variations induced by lighting and viewing conditions, particularly for reflective surfaces and high dynamic range (HDR) environments

Recent advances in neural rendering have demonstrated significant progress in novel view synthesis using neural radiance fields (NeRF) [Mildenhall et al. 2020] and related techniques [Barron et al. 2022; Chen et al. 2022; Kerbl et al. 2023; Verbin et al. 2022; Yu et al. 2021]. However, these methods often struggle with efficient representation and rendering when handling dynamic lighting effects, especially for reflective materials that exhibit complex view-dependent and lighting-dependent appearance. Furthermore, achieving accurate geometric fidelity and sharp boundaries remains challenging, particularly when opacity distributions are ambiguous or poorly constrained. Existing approaches also face difficulties

in HDR reconstruction where bright regions are saturated, losing critical lighting information.

In this paper, we propose a novel scene representation and rendering framework that addresses these challenges by combining 3D Gaussian splatting [Kerbl et al. 2023] with a Ref-NeRF [Verbin et al. 2022] inspired dual spherical harmonics (SH) decomposition. Our method differs from Ref-NeRF in several key ways: while Ref-NeRF uses a single MLP to model both diffuse and specular components, we employ two separate SH representations—one for diffuse and one for specular reflection. Additionally, unlike Ref-NeRF which relies on view direction for specular modelling, we use the incoming light direction to compute the reflection direction, providing more physically accurate specular interactions. Critically, we integrate these advances within the efficient 3D Gaussian splatting framework rather than the computationally expensive volumetric rendering of NeRF. Our approach represents static scenes as a collection of 3D Gaussians, where each Gaussian encodes separate diffuse and specular radiance properties through distinct SH coefficients. The key innovation lies in using different directional parameterization: view direction for diffuse components and reflection direction for specular components, enabling realistic modelling of reflective surfaces.

To address geometric fidelity challenges, we introduce a comprehensive optimization strategy that includes: (1) silhouette-guided Gaussian optimization using binary mask supervision to sharpen opacity boundaries, (2) HDR saturation loss to handle bright regions where ground truth images are clamped, and (3) multi-faceted normal vector optimization with photometric, geometric, and physics-based supervision. These components work together to ensure accurate scene reconstruction with realistic material properties and sharp geometric boundaries. Our contributions can be summarized as follows:

- Dual spherical harmonics decomposition for 3D Gaussian splatting that separately models diffuse and specular reflection components using view direction and reflection direction parameterizations, respectively.
- A comprehensive optimization framework combining: (a) silhouette-guided geometric constraints for sharp opacity boundaries, (b) HDR saturation loss for handling bright regions in HDR environments, and (c) multi-faceted normal vector optimization with photometric, geometric, and physics-based supervision.
- A real-time rendering pipeline that achieves high visual fidelity for dynamic lighting while maintaining computational efficiency suitable for interactive applications.

We validate our approach on the DiLiGenT-MV benchmark dataset [Li et al. 2020], demonstrating superior performance in handling complex lighting interactions and reflective materials. Our results show significant improvements in PSNR, SSIM [Wang et al. 2004], and LPIPS [Zhang et al. 2018] metrics compared to existing methods, while achieving real-time rendering performance. The effectiveness of our dual SH decomposition, silhouette-guided optimization, and normal vector learning is demonstrated through comprehensive ablation studies and visual comparisons.

2 Related Work

2.1 Neural Radiance Fields (NeRF)

Early deep learning approaches for novel view synthesis utilized convolutional neural networks for estimating blending weights [Hedman et al. 2018] or texture-space solutions [Riegler and Koltun 2020; Thies et al. 2019], but suffered from multi-view stereo geometry limitations and temporal inconsistencies.

Volumetric representations marked a significant shift in novel-view synthesis, with Soft3D [Penner and Zhang 2017] laying groundwork for deep-learning driven volumetric ray-marching techniques [Henzler et al. 2019; Sitzmann et al. 2019] that use continuous, differentiable density fields. While Neural Radiance Fields (NeRFs) [Mildenhall et al. 2020] improved rendering quality through importance sampling and positional encoding, their large MLPs significantly impact rendering speed. Despite computational limitations from high volume queries, NeRF inspired extensive follow-up research, with methods like Mip-NeRF360 [Barron et al. 2022] achieving exceptional quality at the cost of extensive training and rendering times.

Recent efforts have focused on balancing quality with faster training and rendering by exploring three primary strategies: spatial data structures for storing neural features, alternative encoding schemes, and reduced MLP complexity. Examples include space discretization techniques [Chen et al. 2022; Fridovich-Keil et al. 2022; Garbin et al. 2021; Hedman et al. 2021; Reiser et al. 2021; Takikawa et al. 2021; Wu et al. 2022; Yu et al. 2021], advanced encoding methods such as hash tables [Müller et al. 2022]. These innovations have enabled smaller MLPs or even eliminated neural networks altogether in some cases [Fridovich-Keil et al. 2022; Sun et al. 2022]. Among these methods, InstantNGP [Müller et al. 2022] uses hash grids with compact MLPs and spherical harmonics for directional effects, while Plenoxels [Fridovich-Keil et al. 2022] employs sparse voxel grids to eliminate neural networks entirely. However, these methods struggle with empty space representation and are limited by structured grids that constrain quality and speed due to high ray-marching sampling demands.

Beyond novel-view synthesis under fixed capture illumination, a growing line of work seeks to make radiance-field representations relightable by disentangling intrinsic scene factors (geometry, reflectance/BRDF, and illumination) or by conditioning the field on explicit lighting parameters. The neural rendering survey by Tewari et al. [Tewari et al. 2020] summarizes this design space and its trade-offs. In practice, relightable radiance fields commonly rely on: (i) inverse rendering-style factorization that predicts surface normals and reflectance while optimizing an incident lighting representation such as low-order spherical harmonics (SH) or HDR environment maps [Ramamoorthi and Hanrahan 2001; Sloan et al. 2002] (ii) reflectance-aware radiance fields that better model view-dependent specular effects to separate material from lighting, e.g., Ref-NeRF [Verbin et al. 2022] and (iii) lighting-conditioned fields that attach SH coefficients or other lighting descriptors to spatial features, enabling re-evaluation under novel illumination [Müller et al. 2022]. While these approaches improve editability and relighting fidelity, they often inherit NeRF’s expensive volumetric sampling and require careful regularization to avoid “baking in” lighting.

2.2 3D Gaussian Splatting (3DGS)

3D Gaussian Splatting (3DGS) [Kerbl et al. 2023] represents a transformative approach to 3D scene representation, leveraging unstructured clouds of ellipsoidal primitives, referred to as “Gaussian splats”. Each splat is defined by its position, color, size, and opacity, enabling highly detailed and photorealistic rendering. Unlike traditional polygon based models or neural network-based methods such as Neural Radiance Fields (NeRF), 3DGS relies on rasterization techniques that are computationally efficient and well-suited for real-time applications such as virtual reality (VR) and augmented reality (AR) environments.

The core innovation of 3DGS lies in its explicit representation using anisotropic 3D Gaussians parameterized by position μ , covariance matrix $\Sigma = RSS^T R^T$ (decomposed into scaling S and rotation R matrices), opacity α , and spherical harmonic coefficients for view-dependent color. This explicit parameterization enables direct gradient-based optimization without neural network inference overhead.

The 3DGS pipeline initializes from SfM point clouds [Schönbberger and Frahm 2016], transforms points into Gaussian splats, and employs adaptive density control through splitting / cloning for under-reconstructed regions and pruning for over-reconstructed areas. Differentiable rasterization projects splats onto 2D planes using efficient GPU-based sorting and alpha-blending.

The rendering process follows a tile-based approach where Gaussians are first culled based on viewing frustum and projected to 2D screen space. The projected 2D Gaussians are then sorted by depth and rasterized using alpha-blending in front-to-back order. This rasterization pipeline achieves real-time performance while maintaining high visual quality, typically rendering at over 100 FPS at 1080p resolution.

Since its introduction, 3DGS has inspired numerous extensions and improvements. Methods such as Mip-Splatting [Yu et al. 2024] address aliasing artifacts through 3D smoothing and 2D Mip mapping, while 4D Gaussian Splatting [Wu et al. 2024] extends the representation to dynamic scenes by incorporating temporal dimensions. Other notable developments include GS-IR [Liang et al. 2023] for relighting applications, and various compression techniques to reduce memory requirements [Bagdasarian et al. 2025]. The explicit nature of 3DGS representations has also facilitated applications in semantic segmentation, object manipulation, and scene editing, demonstrating the versatility of this approach beyond novel view synthesis.

2.3 Limitations of Current Approaches

Despite the significant advances in neural rendering and 3D representation techniques, a critical limitation persists across most state-of-the-art methods: the extracted representations are typically not relightable under novel lighting conditions. While methods like NeRF, 3DGS, and their variants excel at novel view synthesis for fixed lighting scenarios, they fundamentally bake lighting information into the learned representation, making it challenging to separate geometry, material properties, and illumination. Even with methods that are able to edit material properties it is still not possible to dynamically adjust the incident lighting and see its effects on the scene. This limitation severely restricts their applicability

in scenarios requiring dynamic lighting changes, such as virtual production, interactive gaming, and augmented reality applications where lighting conditions must adapt to user interactions or environmental changes. The inability to decompose scenes into intrinsic components geometry, reflectance, and lighting—represents a significant gap in current 3D reconstruction and rendering pipelines, motivating the need for approaches that can produce truly relightable scene representations.

3 Methodology

Our method takes in a set of images of a static scene captured under varying lighting conditions with calibrated camera parameters and known point light source positions. We assume each image is illuminated by a single, distant point light source with known direction and intensity, allowing us to model incident lighting as directional illumination. Under these controlled lighting conditions, we create a set of 3D Gaussians that represent the scene geometry and model the changing outgoing radiance under varying illumination, enabling novel view synthesis under arbitrary point light configurations (Figure 2).

3.1 Method Overview

We formalize our input and representation with set notation:

$$\mathcal{I} = \{I_i\}_{i=1}^{N_I}, \quad \mathcal{C} = \{C_i\}_{i=1}^{N_C}, \quad C_i = (K_i, R_i, t_i), \quad \mathcal{L} = \{\ell_i\}_{i=1}^{N_L} \quad (1)$$

where $\mathcal{I}$ is the set of multi-view images, $\mathcal{C}$ the calibrated camera parameters per view (intrinsics K_i , rotation R_i , translation t_i), and $\mathcal{L}$ the known lighting directions/intensities. We initialize a set of 3D Gaussians

$$\mathcal{G}_0 = \{g_k\}_{k=1}^{N_G}, \quad g_k = (\mu_k, \Sigma_k, \tau_k, \{c_{lm,k}^{\text{diffuse}}\}, \{c_{lm,k}^{\text{specular}}\}, \mathbf{n}_k, \rho_k, \mathbf{s}_k) \quad (2)$$

where $\mu_k \in \mathbb{R}^3$ is the Gaussian center, $\Sigma_k \in \mathbb{R}^{3 \times 3}$ is the covariance matrix, τ_k is the base opacity, $\{c_{lm,k}^{\text{diffuse}}\}$ and $\{c_{lm,k}^{\text{specular}}\}$ are diffuse and specular spherical harmonics coefficients, $\mathbf{n}_k \in \mathbb{S}^2$ is the surface normal, $\rho_k \in [0, 1]$ controls surface roughness, and $\mathbf{s}_k \in [0, 1]^3$ is the specular tint parameter.

Our approach consists of four main components:

- **3D Gaussian Representation:** A scene geometry and appearance model using 3D Gaussian primitives
- **Dual Spherical Harmonics Decomposition:** Separate modelling of diffuse and specular effects
- **Silhouette-Guided Optimization:** A strategy to improve geometric accuracy using binary mask supervision
- **Alpha-Compositing Renderer:** An efficient renderer that combines these elements for real-time visualization

Starting from $\mathcal{I}, \mathcal{C}, \mathcal{L}$, we initialize $\mathcal{G}_0$ and optimize to $\mathcal{G}^*$ using photometric, silhouette, HDR, and normal-related objectives. The result is a compact representation capable of high-quality novel view synthesis under arbitrary lighting.

3.2 3D Gaussian Splatting Foundation

Our method extends the 3D Gaussian Splatting framework, which represents scenes using 3D Gaussian primitives. While the original 3D Gaussian Splatting effectively models static scenes with fixed lighting, our approach introduces several key advances to

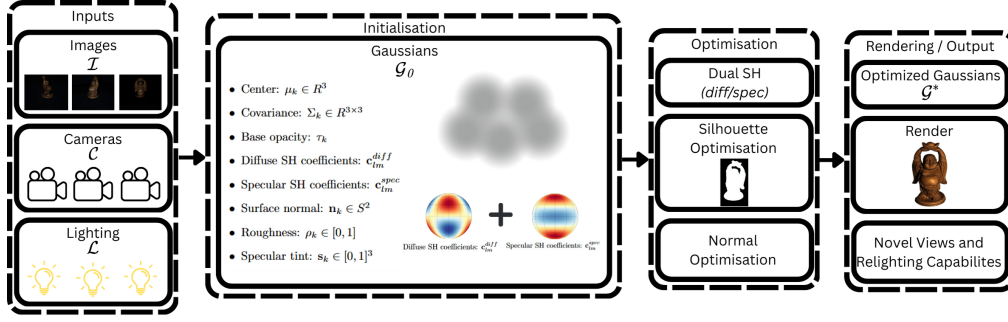


Figure 2: Method overview. Our approach takes multi-view images with known lighting and camera parameters to optimize 3D Gaussians with dual spherical harmonics for diffuse and specular components. Key innovations include silhouette-guided optimization and normal vector learning for realistic relighting and novel view synthesis.

handle dynamic lighting conditions while maintaining its efficient rendering capabilities.

The original rendering equation alpha-composites 2D projected Gaussians to form the pixel color:

$$C = \sum_{i=1}^N c_i^{\text{base}} \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j) \quad (3)$$

where C is the final pixel color, N is the number of contributing Gaussians sorted by depth, c_i^{base} is the base RGB color of the i -th Gaussian, α_i is the opacity after 2D projection, and $\prod_{j=1}^{i-1} (1 - \alpha_j)$ is the accumulated transmittance.

The 3D covariance matrix Σ_i is parameterized using scale $S_i \in \mathbb{R}^{3 \times 3}$ (diagonal matrix) and rotation $R_i \in \mathbb{R}^{3 \times 3}$ (orthogonal matrix):

$$\Sigma_i = R_i S_i S_i^T R_i^T \quad (4)$$

where R_i is a rotation matrix and S_i is a scaling matrix defining the Gaussian's extent.

Optimization proceeds via adaptive density control, which adds Gaussians in high-error regions while pruning insignificant ones; anisotropic scaling that adapts Gaussians to local surface geometry; and color optimization that minimizes the photometric loss between rendered outputs and ground truth images.

Our approach extends this foundation by incorporating a new rendering model and further losses to achieve object modelling with dynamic lighting as detailed in the further sections.

3.3 Surface Property Parameterization

Each Gaussian is augmented with surface properties that control reflection behaviour:

$$\rho_i \in [0, 1] \quad (\text{roughness parameter}) \quad (5)$$

$$s_i \in [0, 1]^3 \quad (\text{specular tint}) \quad (6)$$

$$\mathbf{n}_i \in \mathbb{S}^2 \quad (\text{surface normal}) \quad (7)$$

These parameters enable fine-grained control over material appearance. The roughness parameter ρ_i controls reflection sharpness allowing surfaces to transition smoothly from mirror-like to diffuse reflections. The specular tint s_i enables coloured reflections essential for realistic metallic materials, while the surface normal $\mathbf{n}_i$ determines both reflection direction and lighting interaction, ensuring physically consistent light-surface interactions.

3.4 Lighting-Aware Dual Spherical Harmonics Decomposition

Our core innovation is the decomposition of radiance into diffuse and specular reflection components, using dual spherical harmonics (SH) representations with reflection direction parameterization. This allows our model to effectively handle reflective surfaces while maintaining view-dependent effects.

The final pixel color C_{final} at position p is computed through alpha compositing of multiple Gaussians:

$$C_{\text{final}}(p) = \sum_{i=1}^N \alpha_i T_i \left[c_i^{\text{diffuse}} + s_i \odot c_i^{\text{specular}} \odot I_{\text{light}} \right] + T_{\text{final}} c_{\text{bg}} \quad (8)$$

where $C_{\text{final}}(p)$ is the final rendered color at pixel position p , N represents the number of contributing Gaussians sorted by depth, α_i is the opacity weight of the i -th Gaussian, $T_i = \prod_{j=1}^{i-1} (1 - \alpha_j)$ denotes the accumulated transmittance up to the i -th Gaussian, c_i^{diffuse} is the diffuse color component of the i -th Gaussian, c_i^{specular} is the specular reflection component of the i -th Gaussian, s_i is the specular tint parameter allowing per-Gaussian control over reflection color, I_{light} represents the light intensity as an RGB vector, $\odot$ denotes element-wise multiplication, T_{final} is the remaining transmittance after all Gaussians, and c_{bg} is the background color.

This formulation enables separate modelling of diffuse surface appearance and specular reflections, crucial for realistic rendering of reflective materials under varying lighting conditions.

3.5 Dual Spherical Harmonics with Reflection Direction

The diffuse and specular components are computed through separate spherical harmonics projections using different directional parameterization:

$$c_i^{\text{diffuse}} = \sum_{l=0}^L \sum_{m=-l}^l c_{lm}^{\text{diffuse}} Y_l^m(\omega_{\text{view}}) \quad (9)$$

$$c_i^{\text{specular}} = \text{SH}_{\text{IDE}} \left(c_{lm}^{\text{specular}}, \omega_{\text{reflection}} \cdot \rho_i \right) \quad (10)$$

where c_i^{diffuse} is the diffuse color evaluated using view direction ω_{view} , c_i^{specular} is the specular color evaluated using reflection direction $\omega_{\text{reflection}}$, c_{lm}^{diffuse} and c_{lm}^{specular} are separate sets of SH coefficients, SH_{IDE} denotes spherical harmonics evaluation with Integrated Directional Encoding, ρ_i is the surface roughness parameter controlling reflection sharpness, and $L = 3$ is the maximum SH degree providing sufficient expressivity while maintaining efficiency.

Direction Vector Calculations. : Different directional parameterizations are used for diffuse and specular components:

View Direction:

$$\omega_{\text{view}} = \frac{\mathbf{p}_{\text{cam}} - \boldsymbol{\mu}_i}{\|\mathbf{p}_{\text{cam}} - \boldsymbol{\mu}_i\|_2} \quad (11)$$

Reflection Direction:

$$\omega_{\text{reflection}} = \omega_{\text{light}} - 2(\omega_{\text{light}} \cdot \mathbf{n}_i)\mathbf{n}_i \quad (12)$$

where $\mathbf{p}_{\text{cam}}$ is the camera position, $\boldsymbol{\mu}_i$ is the i -th Gaussian's center position, ω_{light} is the normalized incident light direction, $\mathbf{n}_i$ is the surface normal of the i -th Gaussian, and $\omega_{\text{reflection}}$ follows the law of reflection: $\mathbf{R} = \mathbf{L} - 2(\mathbf{L} \cdot \mathbf{N})\mathbf{N}$.

Integrated Directional Encoding (IDE). For the specular component, we employ Integrated Directional Encoding to account for surface roughness:

$$\text{SH}_{\text{IDE}}(c_{lm}, \omega, \rho) = \sum_{l=0}^L \sum_{m=-l}^l c_{lm} \cdot W_l(\rho) \cdot Y_l^m(\omega) \quad (13)$$

where $W_l(\rho)$ is a roughness-dependent weighting function that attenuates higher-frequency SH terms. Rougher surfaces ($\rho \rightarrow 1$) suppress high-frequency components, creating blurred reflections, while smoother surfaces ($\rho \rightarrow 0$) preserve high-frequency components, producing sharp reflections. This provides a smooth transition from mirror-like to diffuse appearance.

3.6 Rendering Components

The key components of our lighting model interact as follows:

Diffuse Component. : The diffuse component is computed via standard spherical harmonics evaluation along the view direction. This provides view-dependent diffuse appearance that captures how the surface's base color varies with viewing angle.

Specular Reflection Component. : The specular component is evaluated along the reflection direction using Integrated Directional Encoding. The reflection direction parameterization enables proper modelling of mirror-like reflections and specular highlights, ensuring physically accurate rendering of reflective materials.

Surface Roughness Control. : The IDE weighting function modulates reflection sharpness:

$$W_l(\rho) = \exp\left(-\frac{l^2(l+1)^2\rho^2}{2}\right) \quad (14)$$

where smoother surfaces ($\rho \rightarrow 0$) preserve high-frequency SH terms for sharp reflections, while rougher surfaces ($\rho \rightarrow 1$) attenuate them for blurred reflections.

Specular Tinting. : Per-Gaussian specular color control enables realistic metallic materials:

$$c_{\text{final}}^{\text{specular}} = \mathbf{s}_i \odot c_i^{\text{specular}} \quad (15)$$

where $\mathbf{s}_i \in [0, 1]^3$ allows independent control over RGB specular reflection intensity.

Alpha Compositing. : The opacity term remains unchanged from standard Gaussian Splatting:

$$\alpha_i = \min\left(0.99, \tau_i \exp\left(-\frac{1}{2}\mathbf{d}_i^T \tilde{\Sigma}_i^{-1} \mathbf{d}_i\right)\right) \quad (16)$$

where $\mathbf{d}_i$ is the 2D offset from the Gaussian's projected center to the pixel position, $\tilde{\Sigma}_i$ is the 2D projected covariance of Gaussian i , and τ_i is the learnable base opacity parameter.

3.7 Silhouette-Guided Gaussian Optimization

While photometric supervision alone can produce visually pleasing results, it often struggles to accurately reconstruct object boundaries and may produce semi-transparent "floating" Gaussians. To address this limitation, we introduce a silhouette-guided optimization strategy that leverages binary mask supervision to improve geometric fidelity and opacity sharpness.

Silhouette Consistency Loss. : We define a silhouette consistency loss that measures the discrepancy between the rendered alpha channel and the ground truth binary segmentation mask:

$$\mathcal{L}_{\text{sil}} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \|\alpha_{\text{render}}(p) - \alpha_{\text{gt}}(p)\|_1 \quad (17)$$

where $\mathcal{L}_{\text{sil}}$ is the silhouette consistency loss, $\mathcal{P}$ denotes the pixel space (set of all pixels in the image), $|\mathcal{P}|$ is the total number of pixels, $\alpha_{\text{render}}(p)$ is the rendered alpha channel from accumulated opacities at pixel p , $\alpha_{\text{gt}}(p) \in \{0, 1\}$ is the ground truth binary segmentation mask value at pixel p , and $\|\cdot\|_1$ denotes the L1 norm (absolute difference).

The silhouette consistency loss operates through two complementary mechanisms:

Geometric Carving. : Forces Gaussian distributions to contract toward regions where $\alpha_{\text{gt}} = 1$ by penalizing:

$$\frac{\partial \mathcal{L}_{\text{sil}}}{\partial \boldsymbol{\mu}_i} \propto \sum_{p \in \mathcal{P}} (\alpha_{\text{render}}(p) - \alpha_{\text{gt}}(p)) \cdot \frac{\partial \alpha_{\text{render}}(p)}{\partial \boldsymbol{\mu}_i} \quad (18)$$

where $\frac{\partial \mathcal{L}_{\text{sil}}}{\partial \boldsymbol{\mu}_i}$ is the gradient of the silhouette loss with respect to the i -th Gaussian's mean position $\boldsymbol{\mu}_i$, $\alpha_{\text{render}}(p)$ is the rendered alpha value at pixel p , and $\frac{\partial \alpha_{\text{render}}}{\partial \boldsymbol{\mu}_i}$ is how the rendered alpha changes with respect to Gaussian position.

This gradient drives Gaussian means $\boldsymbol{\mu}_i$ toward masked areas, effectively "carving" the scene geometry to match the object silhouettes.

Opacity Sharpening. : Encourages decisive opacity values ($\tau_i \rightarrow 0$ or 1) through L1 pressure:

$$\frac{\partial \mathcal{L}_{\text{sil}}}{\partial \tau_i} = \text{sign}(\alpha_{\text{render}} - \alpha_{\text{gt}}) \cdot T_i \cdot \exp\left(-\mathbf{d}_i^T \tilde{\Sigma}_i^{-1} \mathbf{d}_i\right) \quad (19)$$

where $\frac{\partial \mathcal{L}_{\text{sil}}}{\partial \tau_i}$ is the gradient of the silhouette loss with respect to the i -th Gaussian's opacity parameter τ_i , $\text{sign}(\alpha_{\text{render}} - \alpha_{\text{gt}})$ gives the

direction to adjust opacity (± 1), T_i is the accumulated transmittance as defined earlier, and $\exp(-\mathbf{d}_i^T \Sigma_i^{-1} \mathbf{d}_i)$ is the Gaussian's spatial influence at the current pixel.

This mechanism reduces semi-transparent “halo” artifacts around object boundaries and eliminates “floaters” in empty regions.

Theoretical Motivation. : The silhouette term implements an implicit entropy minimization on the opacity distribution. From an information theory perspective, our approach can be interpreted as seeking a geometry hypothesis that maximizes information content while remaining consistent with observations.

When considering the entropy of the opacity field:

$$H(\alpha) = - \sum_{p \in \mathcal{P}} \left[\alpha(p) \log \alpha(p) + (1 - \alpha(p)) \log(1 - \alpha(p)) \right] \quad (20)$$

The silhouette loss implicitly minimizes this entropy by pushing opacity values toward binary decisions (either 0 or 1), which have lower entropy than intermediate values. This entropy minimization enforces a more decisive decomposition of the scene into occupied and empty regions, reduces ambiguity in the representation making it more interpretable and geometrically accurate, and complements the photometric loss, which alone may tolerate multiple opacity configurations due to view-dependent appearance effects.

From a Bayesian perspective, our approach can be viewed as incorporating a strong prior on the scene geometry through the silhouette masks, then optimizing the posterior probability of the scene given both image observations and silhouette constraints.

3.8 HDR Saturation Loss

A critical challenge in HDR scene reconstruction is handling saturated pixels, where ground truth images are clamped to the $[0, 1]$ range, losing information about true HDR values. To address this limitation, we introduce an HDR saturation loss that encourages the model to predict values beyond the clamped range for bright pixels.

HDR Reconstruction Loss Formulation. : Our HDR saturation loss operates through a three-step process that identifies saturated regions, creates extrapolated targets, and applies asymmetric penalties:

Saturated Pixel Identification. : We first identify pixels that are likely saturated in the ground truth images:

$$\mathcal{M}_{\text{sat}}(p) = \mathbb{I}[\max(R_{\text{gt}}(p), G_{\text{gt}}(p), B_{\text{gt}}(p)) > \tau_{\text{sat}}] \quad (21)$$

where $\mathcal{M}_{\text{sat}}(p) \in \{0, 1\}$ is the binary saturation mask at pixel p , $R_{\text{gt}}(p)$, $G_{\text{gt}}(p)$, $B_{\text{gt}}(p)$ are the ground truth RGB values at pixel p , $\tau_{\text{sat}} = 0.95$ is the saturation threshold, and $\mathbb{I}[\cdot]$ is the indicator function.

Extrapolated Target Generation. : For saturated pixels, we create extrapolated targets that represent plausible HDR values:

$$I_{\text{target}}(p) = \begin{cases} I_{\text{gt}}(p) \cdot \gamma_{\text{ext}} & \text{if } \mathcal{M}_{\text{sat}}(p) = 1 \\ I_{\text{gt}}(p) & \text{otherwise} \end{cases} \quad (22)$$

where $I_{\text{target}}(p)$ is the extrapolated target color at pixel p , $I_{\text{gt}}(p)$ is the ground truth clamped color, and $\gamma_{\text{ext}} = 10$ is the extrapolation factor.

Theoretical Considerations for HDR Parameter Selection. : Ideally, several HDR-related parameters should be determined based on the specific characteristics of each image and scene. The extrapolation factor γ_{ext} should theoretically correspond to the actual degree of saturation present in each image—that is, the true HDR intensity before sensor saturation occurred. Similarly, the saturation threshold τ_{sat} should reflect the specific sensor characteristics and exposure settings used during capture, while the HDR loss weight λ_{HDR} should be calibrated based on the extent and severity of saturation in the dataset.

However, this information is inherently lost during the image acquisition process when pixel values are clamped to the sensor's dynamic range. Consequently, we must rely on heuristic estimation of these values. Our choices of $\gamma_{\text{ext}} = 10$, $\tau_{\text{sat}} = 0.95$, and $\lambda_{\text{HDR}} = 100.0$ represent reasonable approximations based on typical scene luminance ranges and camera sensor characteristics, though the optimal values vary depending on the scene's actual HDR characteristics, the degree of saturation present in the captured images, and the specific imaging conditions.

Asymmetric HDR Loss Application. : The HDR reconstruction loss is applied only to saturated regions with an asymmetric penalty structure:

$$\mathcal{L}_{\text{HDR}} = \lambda_{\text{HDR}} \sum_{p \in \mathcal{P}} \mathcal{M}_{\text{sat}}(p) \cdot \max(0, I_{\text{target}}(p) - I_{\text{render}}(p)) \quad (23)$$

where $\mathcal{L}_{\text{HDR}}$ is the HDR reconstruction loss, $\lambda_{\text{HDR}} = 100.0$ is the HDR loss weight, $I_{\text{render}}(p)$ is the rendered color at pixel p , and $\max(0, \cdot)$ implements the asymmetric penalty (only penalizes under-prediction).

Asymmetric Penalty Rationale. : The asymmetric nature of our HDR loss is crucial for effective HDR reconstruction. By only penalizing under-prediction ($I_{\text{render}} < I_{\text{target}}$), we encourage the model to predict values higher than the clamped ground truth. This design choice is motivated by the facts that over-prediction preserves HDR information that would otherwise be lost, under-prediction leads to unrealistic darkening of bright regions, and the extrapolated targets provide reasonable upper bounds for plausible HDR values.

3.9 Normal Vector Optimization

Accurate surface normal estimation is critical for realistic relighting, as normals directly control how surfaces interact with incident lighting through Fresnel reflectance, specular distribution, and geometric masking terms. We introduce a comprehensive normal optimization strategy that leverages multi-view photometric consistency, geometric constraints, and physics-based supervision.

Normal Representation and Parameterization. : Each 3D Gaussian is augmented with a learnable normal vector $\mathbf{n}_i \in \mathbb{R}^3$. To ensure numerical stability while maintaining differentiability, we employ a dual representation strategy:

$$\mathbf{n}_i^{\text{raw}} \in \mathbb{R}^3 \text{ (storage), } \mathbf{n}_i = \frac{\mathbf{n}_i^{\text{raw}}}{\|\mathbf{n}_i^{\text{raw}}\|_2} \text{ (usage)} \quad (24)$$

where $\mathbf{n}_i^{\text{raw}}$ are the raw normal parameters stored and optimized, $\mathbf{n}_i$ are the unit normal vectors used in rendering computations, and $\|\cdot\|_2$ denotes the L2 norm for normalization.

This parameterization allows gradient flow through raw parameters while ensuring that accessed normals remain geometrically valid unit vectors.

Initialization and Learning Rate Configuration. : Normal vectors are initialized as random unit vectors and optimized entirely through gradient descent:

$$\mathbf{n}_i^{\text{raw}}(0) \sim \mathcal{N}(0, I), \quad \mathbf{n}_i(0) = \frac{\mathbf{n}_i^{\text{raw}}(0)}{\|\mathbf{n}_i^{\text{raw}}(0)\|_2} \quad (25)$$

Normal parameters share the same learning rate as rotation parameters (λ_{rot}) since both represent orientation information requiring similar optimization dynamics.

Multi-Faceted Optimization Strategy. : Our normal optimization operates through several complementary supervision mechanisms.

Depth-Normal Consistency Loss. : To ensure geometric coherence, we introduce a depth-normal consistency constraint:

$$\mathcal{L}_{dn} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \mathcal{W}(p) \|\mathbf{n}_{\text{render}}(p) - \mathbf{n}_{\text{depth}}(p)\|_2^2 \quad (26)$$

where $\mathcal{L}_{dn}$ is the depth-normal consistency loss, $\mathbf{n}_{\text{render}}(p)$ is the rendered normal at pixel p , $\mathbf{n}_{\text{depth}}(p)$ is the normal computed from the depth map gradient, $\mathcal{W}(p)$ is a camera mask weighting function, and $|\mathcal{P}|$ is the total number of pixels.

The depth-derived normals are computed as:

$$\mathbf{n}_{\text{depth}}(p) = \text{normalize} \left(\frac{\partial D(p)}{\partial x} \times \frac{\partial D(p)}{\partial y} \right) \quad (27)$$

where $D(p)$ is the rendered depth at pixel p , and $\times$ denotes the cross product.

Multi-View Consistency Enforcement. : The same normal vector $\mathbf{n}_i$ must produce consistent appearance across all viewpoints. This provides strong supervision through the constraint:

$$\frac{\partial \mathcal{L}_{\text{rgb}}}{\partial \mathbf{n}_i} = \sum_{j=1}^{N_{\text{views}}} \frac{\partial \mathcal{L}_{\text{rgb}}^{(j)}}{\partial \mathbf{n}_i} \quad (28)$$

where N_{views} is the number of training viewpoints and $\mathcal{L}_{\text{rgb}}^{(j)}$ is the photometric loss for viewpoint j .

Densification and Normal Inheritance. : During adaptive densification, child and cloned Gaussians inherit their parent's normal vector: $\mathbf{n}_{\text{child}} = \mathbf{n}_{\text{clone}} = \mathbf{n}_{\text{parent}}$, preserving learned surface orientation while enabling geometric refinement.

3.10 Combined Loss Function and Optimization

We incorporate the silhouette loss, HDR saturation loss, and normal-related losses as objectives with empirically determined weighting:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{rgb}} \mathcal{L}_{\text{rgb}} + \lambda_{\text{sil}} \mathcal{L}_{\text{sil}} + \mathcal{L}_{\text{HDR}} + \lambda_{dn} \mathcal{L}_{dn} \quad (29)$$

where $\mathcal{L}_{\text{total}}$ is the combined loss used for optimization, $\mathcal{L}_{\text{rgb}}$ is the RGB reconstruction loss, $\lambda_{\text{rgb}} = 1.0$ is the weight for the RGB reconstruction loss, $\mathcal{L}_{\text{sil}}$ is the silhouette consistency loss as defined above, $\lambda_{\text{sil}} = 0.5$ is the empirically determined weight for the silhouette consistency loss, $\mathcal{L}_{\text{HDR}}$ is the HDR saturation loss as defined above, $\mathcal{L}_{dn}$ is the depth-normal consistency loss as defined above, and $\lambda_{dn} = 1.0$ is the weight for the depth-normal consistency loss.

The HDR saturation loss is integrated into our overall optimization objective to ensure that HDR reconstruction is prioritized for saturated regions while maintaining overall photometric accuracy and geometric consistency. The depth-normal consistency loss ensures that the learned normal vectors are geometrically coherent with the rendered depth information. This comprehensive optimization strategy enables our method to learn physically plausible surface orientations that support realistic relighting under novel illumination conditions while maintaining high-quality novel view synthesis.

The specific value of $\lambda_{\text{sil}} = 0.5$ was determined empirically through experimentation. A higher weight ($\lambda_{\text{sil}} > 0.5$) tends to overemphasize geometric constraints at the expense of appearance quality, while a lower weight ($\lambda_{\text{sil}} < 0.5$) reduces the beneficial effect on geometry.

To ensure that silhouette gradients do not contaminate appearance learning, we implement gradient stopping: silhouette gradients do not backpropagate through color channels. This separation allows the geometric optimization to proceed without disrupting the learning of view and lighting-dependent appearance.

4 Results and Evaluation

We implement our method on top of a 3D Gaussian splatting [Kerbl et al. 2023] renderer enhanced with dual SH heads (diffuse/specular), silhouette and HDR losses, and normal optimization, and we train it using the Adam optimizer with adaptive densification and early stopping criteria. We evaluate on the DiLiGenT-MV [Li et al. 2020] dataset, which provides multi-view images of five objects (BEAR, BUDDHA, COW, POT2, READING) captured under known lighting conditions with calibrated cameras. This dataset is particularly suitable for our evaluation as it includes objects with varying material properties from diffuse to specular. For comparison, we consider seven representative methods: NeRF [Mildenhall et al. 2020], KB22 [Kaya et al. 2021], UNISURF [Oechsle et al. 2021], NeRFactor [Zhang et al. 2021b], NeRD [Boss et al. 2021], PhySG [Zhang et al. 2021a], and PSNerf [Yang et al. 2022]. For quantitative evaluation, we use standard image quality metrics: PSNR (higher is better), SSIM (higher is better), and LPIPS (lower is better).

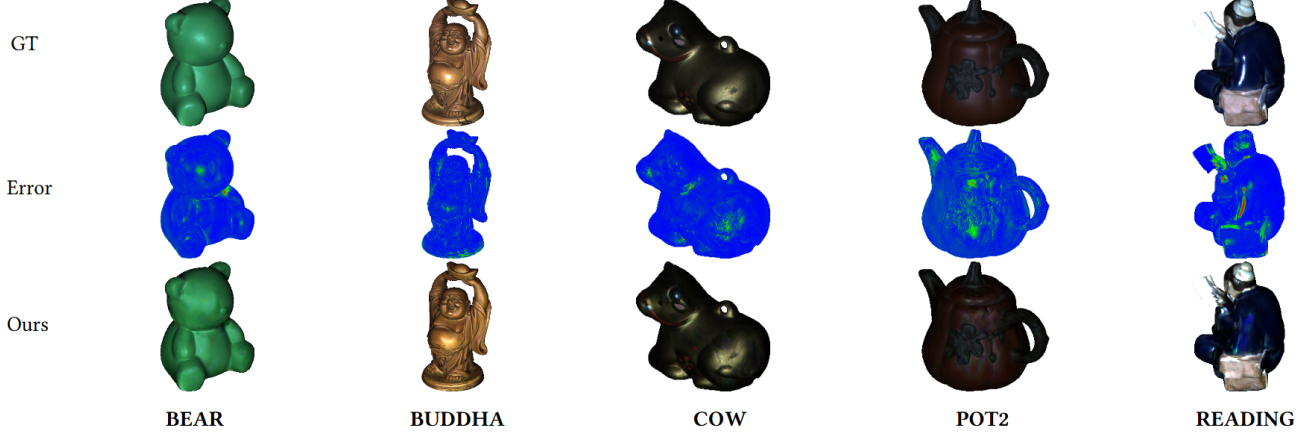
4.1 Quantitative Comparisons

We evaluate our method against state-of-the-art baselines on DiLiGenT-MV [Li et al. 2020] using PSNR, SSIM, and LPIPS metrics. Table 1 shows that our dual SH approach achieves top or competitive scores across all objects, with visual comparisons presented in Figure 3. Overall, our method achieves competitive performance across all metrics: specifically, we obtain the highest PSNR on 3 out of 5 objects and the highest SSIM with the lowest LPIPS on 4 out of 5 objects. This demonstrates the effectiveness of our dual spherical harmonics approach in capturing both diffuse and specular reflections across diverse material properties in the DiLiGenT-MV [Li et al. 2020] dataset.

While PSNerf achieves competitive PSNR and SSIM scores, particularly excelling on COW (42.39 dB PSNR) and READING (30.47 dB PSNR), our method significantly outperforms it in LPIPS across most objects and achieves superior SSIM consistency. Specifically, our LPIPS improvements range from 0.03 to 0.5 across objects, with the

Table 1: Comparison of objects in the DiLiGenT-MV [Li et al. 2020] dataset. The best values for each metric are shown in bold.

Method	BEAR			BUDDHA			COW			POT2			READING		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
NeRF [Mildenhall et al. 2020]	30.98	0.9887	1.09	29.31	0.9664	2.8	33.03	0.9907	0.54	32.52	0.9842	1.41	26.58	0.9664	1.84
KB22 [Kaya et al. 2021]	39.63	0.9960	0.24	33.62	0.9844	0.56	31.38	0.9800	0.87	33.39	0.9767	1.02	22.45	0.9560	2.11
UNISURF [Oechsle et al. 2021]	40.13	0.9954	0.4	30.98	0.9707	1.98	40.17	0.9953	0.39	43.06	0.9954	0.5	22.19	0.9579	2.29
NeRFactor [Zhang et al. 2021b]	29.28	0.9791	2.38	26.34	0.9385	6.33	27.60	0.9630	1.67	32.32	0.9738	1.84	25.62	0.9468	2.99
NeRD [Boss et al. 2021]	26.24	0.9661	3.56	20.94	0.8701	6.25	23.98	0.8914	1.97	26.34	0.8422	2.19	20.13	0.9023	3.9
PhySG [Zhang et al. 2021a]	34.01	0.9841	1.6	29.64	0.9594	2.65	34.38	0.9856	1.02	35.92	0.9814	1.23	24.19	0.9531	2.88
PSNerf [Yang et al. 2022]	41.58	0.9959	0.31	33.73	0.9829	0.54	42.39	0.9962	0.22	45.44	0.9960	0.15	30.47	0.9808	0.75
Ours	42.36	0.9958	0.17	37.24	0.9901	0.16	41.00	0.9967	0.18	46.65	0.9969	0.18	24.70	0.9868	0.25

**Figure 3: Visual comparison on DiLiGenT-MV [Li et al. 2020] dataset. Top: ground truth. Middle: error maps. Bottom: our results.****Table 2: Ablation Study Results**

Method	BEAR			BUDDHA			COW			POT2			READING		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	42.36	0.9958	0.17	37.24	0.9901	0.16	41.00	0.9967	0.18	46.65	0.9969	0.18	24.70	0.9868	0.25
w/o $\mathcal{L}_{HDR}$	41.97	0.9954	0.17	37.08	0.9898	0.15	40.62	0.9967	0.18	46.09	0.9969	0.18	27.43	0.9911	0.24
w/o Normals	42.22	0.9957	0.17	37.26	0.9900	0.13	41.07	0.9967	0.18	45.80	0.9963	0.18	24.92	0.9871	0.25
$\lambda_{sil} = 0$	43.52	0.9960	0.17	37.80	0.9897	0.15	40.36	0.9960	0.22	44.92	0.9956	0.23	23.25	0.9771	0.32
$\lambda_{sil} = 1.0$	41.38	0.9953	0.17	36.72	0.9890	0.15	39.37	0.9958	0.22	45.18	0.9952	0.23	24.70	0.9866	0.32

most significant gains on BEAR and READING. Our improvement stems from material-aware modeling and geometric constraints, which better preserve perceptual quality and fine details. The dual SH representation allows for more accurate separation of diffuse and specular components, leading to sharper material boundaries and more realistic highlight rendering compared to PSNerf.

Turning to volumetric methods such as neural radiance field variants (NeRF, NeRFactor, NeRD) consistently underperform across all metrics—NeRF shows the largest deficiency across objects, while NeRFactor and NeRD achieve slightly better results but still fall significantly short. Volumetric MLPs struggle to capture the explicit geometry and material separation that our Gaussian-based approach provides, leading to blurred boundaries and poor material decomposition. The implicit nature of volumetric representations makes it challenging to enforce geometric constraints and results in inconsistent material properties across different viewing angles; additionally, these methods require substantially longer training

times (8–24 hours per object) and cannot achieve real-time rendering performance. Next, surface-based methods such as UNISURF and PhySG show mixed results and particularly struggle with reflective surfaces. UNISURF performs reasonably on diffuse objects like BEAR but struggles on specular surfaces like READING, while PhySG’s physics-based approach shows promise, achieving moderate performance across most objects but struggling with complex material interactions. Our dual SH representation handles both diffuse and specular components more robustly than these surface-based approaches, which often assume simplified material models and cannot adapt to the full range of material properties present in real-world objects; the explicit Gaussian representation also provides better geometric control and, combined with our silhouette loss, reduces artifacts around object boundaries unlike surface-based methods. Finally, looking at efficiency analysis beyond image quality, our method maintains real-time rendering performance while delivering superior results—we achieve over

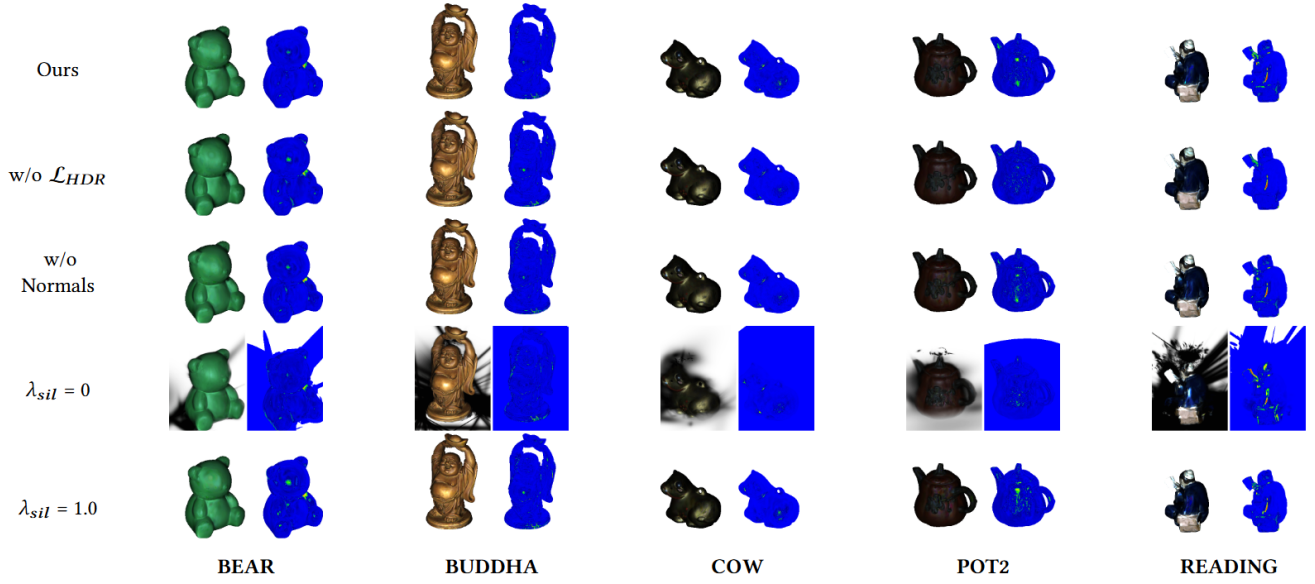


Figure 4: Qualitative ablation study results (prediction | error). Blue indicates low error, red indicates high error.

30 FPS at 1024x1024 resolution, whereas many neural baselines require seconds per frame (orders of magnitude slower); our training times are approximately one hour per object, compared to several hours or even days required by volumetric neural methods like NeRF variants (10–100x faster training).

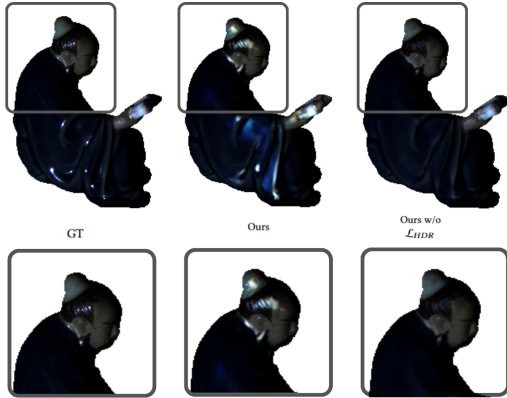


Figure 5: HDR saturation loss contribution demonstration.

4.2 Ablation Studies

We conduct systematic ablation studies to validate the effectiveness of each component in our method. Table 2 presents quantitative results across all test objects, while Figure 4 provides visual comparisons with error maps. We provide detailed analysis of each component’s contribution below.

The HDR saturation loss ($\mathcal{L}_{HDR}$) demonstrates targeted benefits for reflective surfaces rather than universal improvements; for most objects with limited saturation, the impact is modest (PSNR

drops of 0.16–0.56 dB when removed). However, **READING** reveals its specialized function: while removing HDR loss improves PSNR (24.70 to 27.43 dB), visual quality degrades substantially as shown in Figure 5, and the method fails to capture essential specular highlights on reflective surfaces without this constraint. Moreover, eliminating learned surface normals results in moderate performance degradation with object-dependent impact—**BEAR** shows minimal degradation (0.14 dB), while **POT2** experiences substantial drops (0.85 dB). Finally, the silhouette loss addresses specific geometric artifacts rather than general quality improvements: without silhouette guidance ($\lambda_{sil} = 0$), some objects show PSNR improvements (**BEAR**: +1.16 dB, **BUDDHA**: +0.56 dB), but **COW**, **POT2**, and **READING** show LPIPS degradation; the constraint prevents floating geometry artifacts when semi-transparent Gaussians accidentally align with camera viewpoints, and visual inspection confirms that models without silhouette guidance suffer from floating geometry artifacts, less well-defined boundaries, and “halo” effects around object silhouettes.

Object-Specific Analysis of the ablation results reveal how component importance varies with object material properties:

- **BEAR**: Shows unexpected PSNR improvements without silhouette loss (+1.16 dB), yet suffers slightly in SSIM. This demonstrates that silhouette loss can effectively remove unwanted floating geometry artifacts without causing significant degradation.
- **POT2**: Achieves the highest baseline performance (46.65 dB PSNR) and shows LPIPS and SSIM degradation when silhouette loss is removed, confirming that silhouette guidance is crucial for geometric objects with clear boundaries.
- **READING**: Demonstrates targeted HDR loss benefits: removing HDR loss improves PSNR but degrades visual quality by failing to capture essential specular highlights on reflective surfaces.

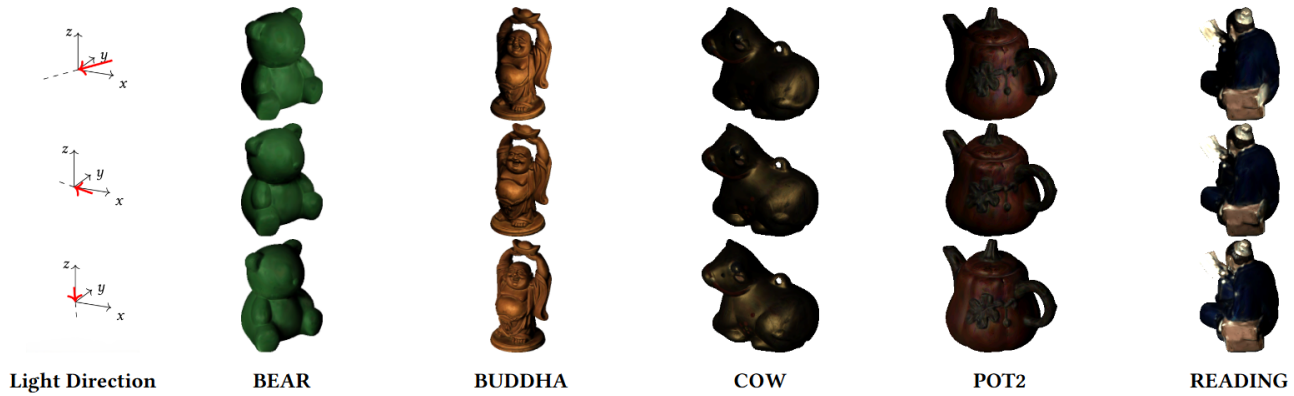


Figure 6: Novel view synthesis under varying illumination. Each row shows a different lighting condition (left column) applied to all five objects. The light direction diagrams show the incident illumination vectors (red arrows) for each lighting condition from top to bottom.

4.3 Qualitative Evaluation: Novel Lighting

Our method’s dual spherical harmonics representation enables robust relighting capabilities under novel illumination conditions. Figure 6 demonstrates qualitative results across four representative objects (BUDDHA, COW, BEAR, POT2) rendered under three distinct lighting directions from the same viewpoint.

4.4 Key Insights and Discussion

Our comprehensive evaluation reveals several important insights about the proposed dual spherical harmonics approach. Our results indicate that each component addresses specific challenges rather than providing universal improvements: for instance, HDR loss offers targeted benefits for reflective surfaces, whereas silhouette guidance specifically addresses multi-view consistency by preventing floating geometry artifacts; accordingly, our error analysis shows that removing silhouette guidance increases error while removing HDR loss has minimal impact except for highly reflective objects. Building on this, component importance varies significantly with object material properties—objects with clear geometric boundaries benefit more from silhouette guidance, while reflective surfaces require HDR loss for accurate highlight capture—suggesting potential for adaptive component weighting based on material analysis. At the same time, our method achieves an excellent balance between rendering efficiency and visual quality, maintaining real-time performance while delivering superior perceptual results compared to existing neural approaches; in particular, the dual SH representation provides computational efficiency without sacrificing the ability to model complex material interactions.

4.5 Limitations

Our approach has limitations that constrain its applicability: (i) the silhouette loss requires clear object boundaries and cannot handle unbounded scenes, (ii) the lighting model supports only single point light sources and struggles with high-frequency effects like sharp shadows or caustics, and (iii) the method is restricted to static scenes without dynamic objects or temporal variations.

4.6 Conclusion and Future Work

We presented a novel framework combining 3D Gaussian splatting with dual spherical harmonics decomposition for high-quality novel view synthesis under arbitrary lighting. Our key innovation separately models diffuse and specular reflection using distinct directional parameterizations: view direction for diffuse and reflection direction for specular components. This dual SH decomposition enables realistic rendering of reflective surfaces while maintaining computational efficiency. Our comprehensive optimization incorporates silhouette-guided geometric constraints for sharp opacity boundaries, HDR saturation loss for bright regions, and multi-faceted normal vector optimization with photometric and physics-based supervision. These innovations achieve superior geometric reconstruction, realistic lighting effects, and enhanced material modeling for real-time applications. Experimental validation on DiLiGenT-MV demonstrates consistent improvements over state-of-the-art approaches in PSNR, SSIM, and LPIPS metrics while achieving fast optimization and real-time rendering. Ablation studies and error map analysis confirm the targeted effectiveness of each component. Our dual spherical harmonics approach advances neural rendering by providing robust, efficient novel view synthesis with dynamic lighting and reflective materials. Future work could explore: (i) extending to unbounded scenes with hierarchical Gaussian representations, (ii) incorporating enhanced lighting models with higher-frequency SH and multiple light sources, and (iii) handling dynamic scenes with temporal coherence constraints and deformation fields.

Acknowledgments

This research was supported by the UKRI EPSRC and BBC Prosperity Partnership AI4ME: Future Personalised Object-Based Media Experiences Delivered at Scale Anywhere EP/V038087. For the purpose of Open Access, the author has applied a Creative Commons Attribution (CC BY) public copyright licence to any Author Accepted Manuscript version arising from this submission.

References

- M. T. Bagdasarian, P. Knoll, Y. Li, F. Barthel, A. Hilsmann, P. Eisert, and W. Morgenstern. 2025. 3DGS.zip: A Survey on 3D Gaussian Splatting Compression Methods. *Computer Graphics Forum* (2025), e70078. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1111/cgf.70078 doi:10.1111/cgf.70078 https://w-m.github.io/3dgs-compression-survey/.
- Jonathan T. Barron, Ben Mildenhall, Dor Verbin, Pratul P. Srinivasan, and Peter Hedman. 2022. Mip-NeRF 360: Unbounded Anti-Aliased Neural Radiance Fields. *CVPR* (2022).
- Mark Boss, Raphael Braun, Varun Jampani, Jonathan T. Barron, Ce Liu, and Hendrik P.A. Lensch. 2021. NeRD: Neural Reflectance Decomposition from Image Collections. In *IEEE International Conference on Computer Vision (ICCV)*.
- Anpei Chen, Zexiang Xu, Andreas Geiger, Jingyi Yu, and Hao Su. 2022. TensorRF: Tensorial Radiance Fields. In *European Conference on Computer Vision (ECCV)*.
- Sara Fridovich-Keil, Alex Yu, Matthew Tancik, Qinhong Chen, Benjamin Recht, and Angjoo Kanazawa. 2022. Plenoxels: Radiance Fields without Neural Networks. In *CVPR*.
- Stephan J. Garbin, Marek Kowalski, Matthew Johnson, Jamie Shotton, and Julien Valentin. 2021. FastNeRF: High-Fidelity Neural Rendering at 200FPS. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*. 14326–14335. doi:10.1109/ICCV48922.2021.01408
- Peter Hedman, Julien Philip, True Price, Jan-Michael Frahm, George Drettakis, and Gabriel Brostow. 2018. Deep blending for free-viewpoint image-based rendering. *ACM Trans. Graph.* 37, 6, Article 257 (Dec. 2018), 15 pages. doi:10.1145/3272127.3275084
- Peter Hedman, Pratul P. Srinivasan, Ben Mildenhall, Jonathan T. Barron, and Paul Debevec. 2021. Baking Neural Radiance Fields for Real-Time View Synthesis. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*. 5855–5864. doi:10.1109/ICCV48922.2021.00582
- Philipp Henzler, Niloy Mitra, and Tobias Ritschel. 2019. Escaping Plato's Cave: 3D Shape From Adversarial Rendering. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*. 9983–9992. doi:10.1109/ICCV.2019.01008
- Berk Kaya, Suryansh Kumar, Francesco Sarno, Vittorio Ferrari, and Luc Van Gool. 2021. Neural Radiance Fields Approach to Deep Multi-View Photometric Stereo. arXiv:2110.05594 [cs.CV] https://arxiv.org/abs/2110.05594
- Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 2023. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics* 42, 4 (July 2023). https://repo-sam.inria.fr/fungraph/3d-gaussian-splatting/
- Min Li, Zhenglong Zhou, Zhe Wu, Boxin Shi, Changyu Diao, and Ping Tan. 2020. Multi-View Photometric Stereo: A Robust Solution and Benchmark Dataset for Spatially Varying Isotropic Materials. *Trans. Img. Proc.* 29 (Jan. 2020), 4159–4173. doi:10.1109/TIP.2020.2968818
- Zhihao Liang, Qi Zhang, Ying Feng, Ying Shan, and Kui Jia. 2023. Gs-ir: 3d gaussian splatting for inverse rendering. *arXiv preprint arXiv:2311.16473* (2023).
- Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. 2020. NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis. In *ECCV*.
- Thomas Müller, Alex Evans, Christoph Schied, and Alexander Keller. 2022. Instant Neural Graphics Primitives with a Multiresolution Hash Encoding. *ACM Trans. Graph.* 41, 4, Article 102 (July 2022), 15 pages. doi:10.1145/3528223.3530127
- Michael Oechsle, Songyou Peng, and Andreas Geiger. 2021. UNISURF: Unifying Neural Implicit Surfaces and Radiance Fields for Multi-View Reconstruction. In *International Conference on Computer Vision (ICCV)*.
- Eric Penner and Li Zhang. 2017. Soft 3D Reconstruction for View Synthesis. 36, 6 (2017).
- Ravi Ramamoorthi and Pat Hanrahan. 2001. An efficient representation for irradiance environment maps. In *Proceedings of the 28th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH '01)*. Association for Computing Machinery, New York, NY, USA, 497–500. doi:10.1145/383259.383317
- Christian Reiser, Songyou Peng, Yiyi Liao, and Andreas Geiger. 2021. KiloNeRF: Speeding up Neural Radiance Fields with Thousands of Tiny MLPs. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*. 14315–14325. doi:10.1109/ICCV48922.2021.01407
- Gernot Riegler and Vladlen Koltun. 2020. Free View Synthesis. In *European Conference on Computer Vision*.
- Johannes L. Schönberger and Jan-Michael Frahm. 2016. Structure-from-Motion Revisited. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 4104–4113. doi:10.1109/CVPR.2016.445
- Vincent Sitzmann, Michael Zollhöfer, and Gordon Wetzstein. 2019. Scene Representation Networks: Continuous 3D-Structure-Aware Neural Scene Representations. In *Advances in Neural Information Processing Systems*.
- Peter-Pike Sloan, Jan Kautz, and John Snyder. 2002. Precomputed radiance transfer for real-time rendering in dynamic, low-frequency lighting environments. *ACM Trans. Graph.* 21, 3 (July 2002), 527–536. doi:10.1145/566654.566612
- Cheng Sun, Min Sun, and Hwann-Tzong Chen. 2022. Direct Voxel Grid Optimization: Super-fast Convergence for Radiance Fields Reconstruction. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 5449–5459. doi:10.1109/CVPR52688.2022.00538
- Towaki Takikawa, Joey Litalien, Kangxue Yin, Karsten Kreis, Charles Loop, Derek Nowrouzezahrai, Alec Jacobson, Morgan McGuire, and Sanja Fidler. 2021. Neural Geometric Level of Detail: Real-time Rendering with Implicit 3D Shapes. (2021).
- A. Tewari, O. Fried, J. Thies, V. Sitzmann, S. Lombardi, K. Sunkavalli, R. Martin-Brualla, T. Simon, J. Saragih, M. Nießner, R. Pandey, S. Fanello, G. Wetzstein, J.-Y. Zhu, C. Theobalt, M. Agrawala, E. Shechtman, D. B. Goldman, and M. Zollhöfer. 2020. State of the Art on Neural Rendering. *Computer Graphics Forum* 39, 2 (2020), 701–727. arXiv:https://onlinelibrary.wiley.com/doi/pdf/10.1111/cgf.14022 doi:10.1111/cgf.14022
- Justus Thies, Michael Zollhöfer, and Matthias Nießner. 2019. Deferred neural rendering: image synthesis using neural textures. *ACM Trans. Graph.* 38, 4, Article 66 (July 2019), 12 pages. doi:10.1145/3306346.3323035
- Dor Verbin, Peter Hedman, Ben Mildenhall, Todd Zickler, Jonathan T. Barron, and Pratul P. Srinivasan. 2022. Ref-NeRF: Structured View-Dependent Appearance for Neural Radiance Fields. *CVPR* (2022).
- Zhou Wang, A.C. Bovik, H.R. Sheikh, and E.P. Simoncelli. 2004. Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing* 13, 4 (2004), 600–612. doi:10.1109/TIP.2003.819861
- Guanjun Wu, Taoran Yi, Jiemin Fang, Lingxi Xie, Xiaopeng Zhang, Wei Wei, Wenyu Liu, Qi Tian, and Xinggang Wang. 2024. 4D Gaussian Splatting for Real-Time Dynamic Scene Rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 20310–20320.
- Tong Wu, Jiaqi Wang, Xingang Pan, Xudong Xu, Christian Theobalt, Ziwei Liu, and Dahua Lin. 2022. Voxurf: Voxel-based Efficient and Accurate Neural Surface Reconstruction. In *International Conference on Learning Representations (ICLR)*.
- Wenqi Yang, Guanying Chen, Chaofeng Chen, Zhenfang Chen, and Kwan-Yee K. Wong. 2022. PS-NeRF: Neural Inverse Rendering for Multi-view Photometric Stereo. In *European Conference on Computer Vision (ECCV)*.
- Alex Yu, Ruilong Li, Matthew Tancik, Hao Li, Ren Ng, and Angjoo Kanazawa. 2021. PlenOctrees for Real-time Rendering of Neural Radiance Fields. In *ICCV*.
- Zehao Yu, Anpei Chen, Binbin Huang, Torsten Sattler, and Andreas Geiger. 2024. Mip-Splatting: Alias-free 3D Gaussian Splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 19447–19456.
- Kai Zhang, Fujun Luan, Qianqian Wang, Kavita Bala, and Noah Snavely. 2021a. PhysSG: Inverse Rendering with Spherical Gaussians for Physics-based Material Editing and Relighting. In *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. 2018. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 586–595. doi:10.1109/CVPR.2018.00068
- Xiuming Zhang, Pratul P. Srinivasan, Boyang Deng, Paul Debevec, William T. Freeman, and Jonathan T. Barron. 2021b. NeRFactor: neural factorization of shape and reflectance under an unknown illumination. *ACM Trans. Graph.* 40, 6, Article 237 (Dec. 2021), 18 pages. doi:10.1145/3478513.3480496